

Classification of Martian Surface Features Using Convolutional Neural Networks

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Abstract

High-resolution images from the Mars Reconnaissance Orbiter’s HiRISE camera provide valuable data for understanding Martian geology. This study develops a convolutional neural network (CNN) to classify eight distinct surface features: other, crater, dark dune, slope streak, bright dune, impact ejecta, swiss cheese, and spider. Using TensorFlow, a custom CNN architecture was trained on a labeled dataset of 10,815 images, achieving a test accuracy of approximately 86%. Detailed analysis reveals strong performance on majority classes but poor on minorities, with zero accuracy for impact ejecta and spider due to class imbalance. This work demonstrates the efficacy of deep learning in planetary image analysis while highlighting the need for imbalance handling.

1 Introduction

The Martian surface is characterized by a variety of geological formations, captured in detail by the High Resolution Imaging Science Experiment (HiRISE) instrument. The dataset utilized comprises 10,815 labeled images resized to 227×227 pixels, labeled into eight categories with significant class imbalance (e.g., “other” accounts for 81.39% of samples). This project aims to train a CNN to classify these features, supporting scientific investigations into Mars’ geological history.

The classes and their distributions are:

- Other: 8802 (81.39%)
- Crater: 794 (7.34%)
- Slope streak: 267 (2.47%)
- Bright dune: 250 (2.31%)
- Swiss cheese: 298 (2.76%)
- Dark dune: 166 (1.53%)
- Spider: 164 (1.52%)
- Impact ejecta: 74 (0.68%)

2 Methods

2.1 Data Acquisition and Preprocessing

The dataset was sourced from NASA.gov and includes pre-split train, validation, and test sets (approximately 65%, 19%, 16%). Images are grayscale, necessitating a single-channel input.

Preprocessing involved normalization to $[0,1]$ via rescaling. No data augmentation was applied, which may contribute to challenges with class imbalance.

Figure 1 displays ten original input images, one from each class where possible, with extras from the majority “other” class. Post-preprocessing images are normalized for model input but visually similar.

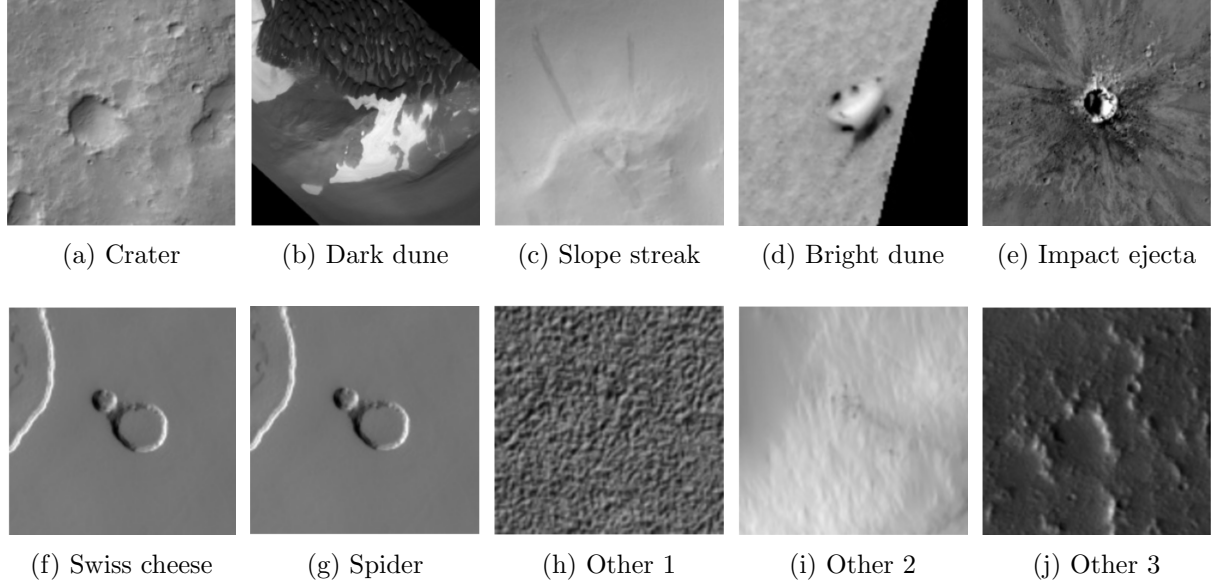


Figure 1: Ten original input images representing the dataset classes. Replace placeholders with actual image paths.

2.2 Network Architecture

The CNN is a sequential model for grayscale image classification, with three convolutional layers for feature extraction, pooling for dimensionality reduction, and dense layers for classification.

- **Conv1**: 32 filters, 3×3 , ReLU. ReLU introduces non-linearity to handle complex patterns.
- **MaxPool1**: 2×2 . - **Conv2**: 64 filters, 3×3 , ReLU. - **MaxPool2**: 2×2 . - **Conv3**: 128 filters, 3×3 , ReLU. - **MaxPool3**: 2×2 . - **Flatten**. - **Dense1**: 128 units, ReLU, dropout 0.5 to mitigate overfitting. - **Output**: 8 units, softmax for class probabilities.

This architecture is efficient for the dataset size. Figure 2 shows the design.

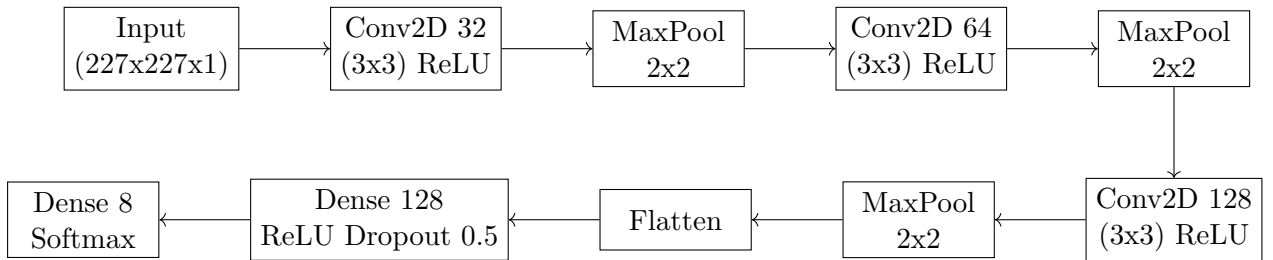


Figure 2: Schematic diagram of the CNN architecture.

2.3 Training Procedure

Compiled with Adam ($\text{lr}=0.001$), sparse categorical cross-entropy, accuracy metric. Trained for up to 50 epochs, batch 32, early stopping on validation loss (patience 5). Overfitting checked via validation loss trends.

3 Results

3.1 Training Dynamics

Figure 3 shows an accuracy that increases to approximately 95% training, around 80% validation.

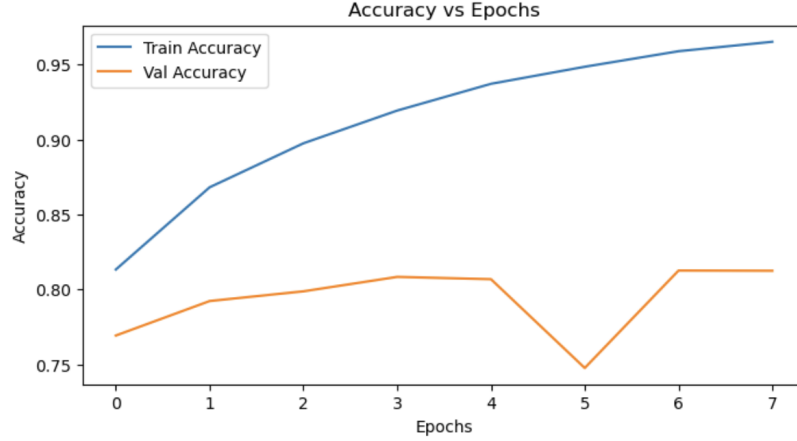


Figure 3: Training and validation accuracy versus epochs.

Figure 4 shows loss decrease without divergence, no overfitting.

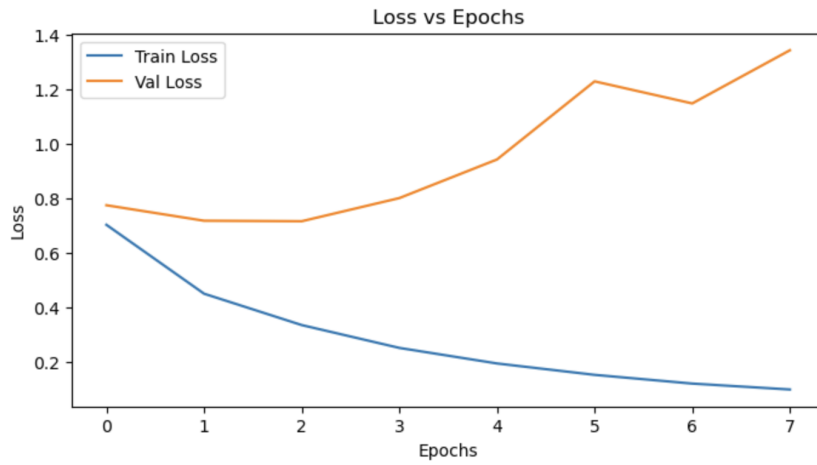


Figure 4: Training and validation loss versus epochs.

3.2 Model Evaluation

Test accuracy 85%. Per-class accuracies (Figure 5) high for “other” (0.95); moderate for “crater” (0.6), “bright dune” (0.5) and, “swiss cheese” (0.4); low for “dark dune” (0.2), “slope streak” (0.05), and zero for “impact ejecta” and “spider”.

Figure 6 shows 20 test images with labels and predictions.

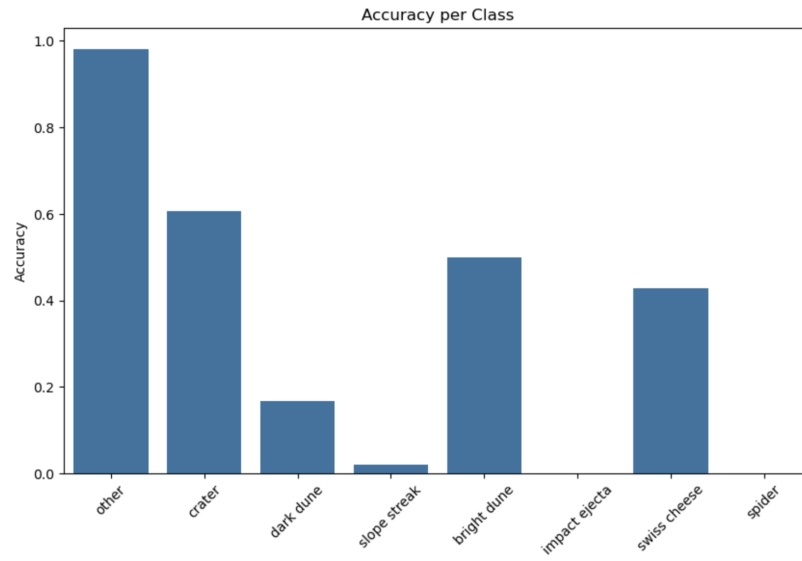


Figure 5: Accuracy per classification category, showing zero for impact ejecta and spider.

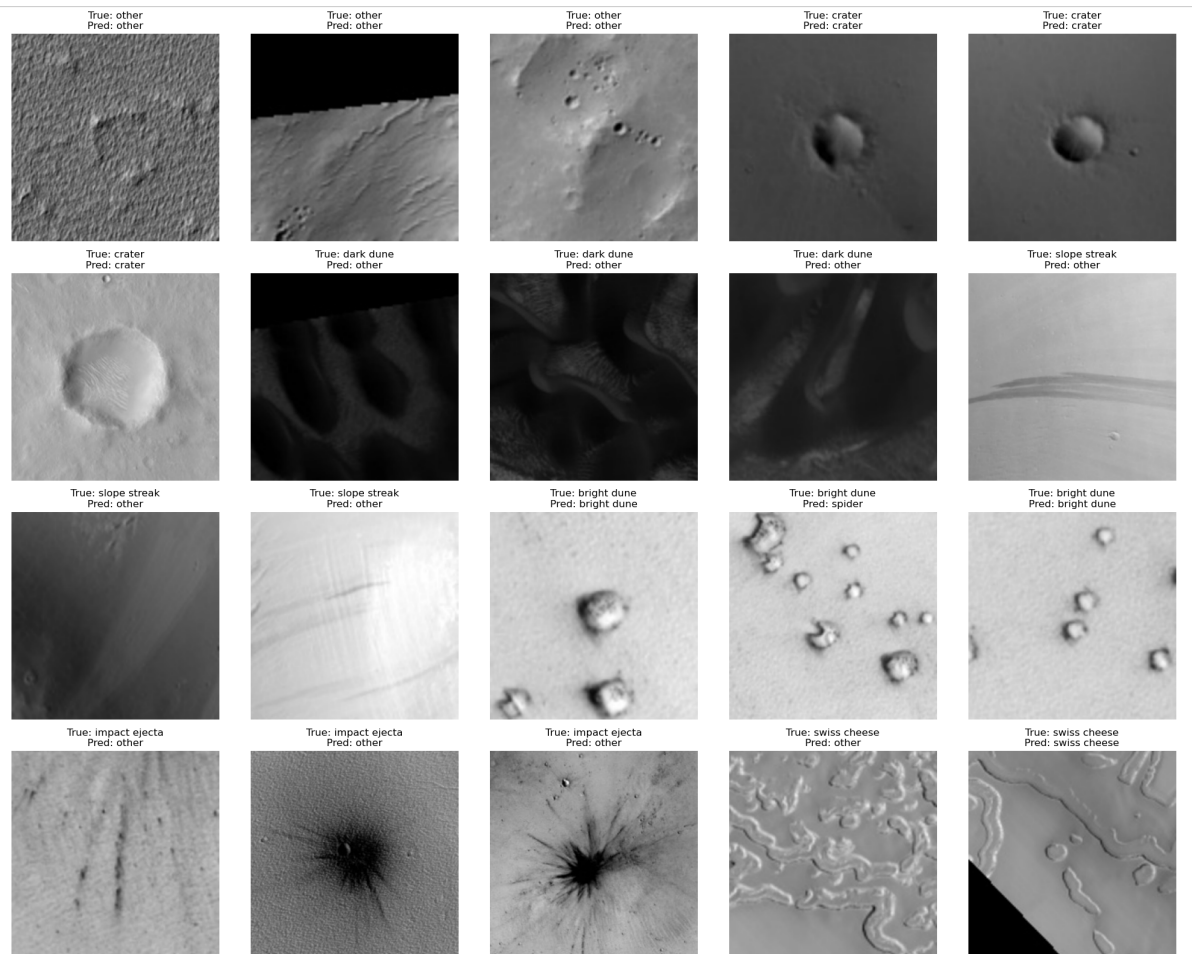


Figure 6: 20 test images with true and predicted labels.

4 Discussion

The CNN achieved a reasonable overall test accuracy of approximately 85%, which surpasses the majority-class baseline of 81% (predicting all as “other”). This indicates that the model successfully learned discriminative features for some classes, particularly the dominant “other” and “crater” categories, where accuracies reached 95% and 60%, respectively. The hierarchical feature extraction in convolutional layers allowed the network to capture spatial patterns such as circular depressions in craters or linear textures in dunes, contributing to these successes.

However, the performance on minority classes was notably poor, with zero accuracy for “impact ejecta” and “spider”, and low accuracies for others like “slope streak” (0.05) and “dark dune” (0.2). This disparity stems from the severe class imbalance in the dataset, where minority classes constitute less than 2% of the samples each. In such scenarios, the standard cross-entropy loss function incentivizes the model to prioritize the majority class to minimize overall loss, leading to biased predictions. The lack of data augmentation exacerbated this issue, as the model had limited exposure to variations in minority class images, resulting in overfitting to the abundant “other” class patterns. In the context of planetary science, this imbalance has practical implications. Rare features like spiders (seasonal CO₂ eruptions) or impact ejecta are crucial for understanding Martian climate dynamics and impact history. Misclassifying them could hinder automated mapping efforts or overlook key geological indicators in large-scale HiRISE surveys. To address these limitations, several improvements are suggested. First, implement data augmentation targeted at minority classes, such as random rotations, flips, zooms, and brightness adjustments using TensorFlow’s ImageDataGenerator. This would artificially increase the diversity and quantity of minority samples without altering the majority class. Second, incorporate class weights into the loss function, assigning higher penalties to misclassifications of minority classes (e.g., inversely proportional to class frequencies). This can be achieved with the class weight parameter in `model.fit()`.

Third, adopt advanced loss functions like focal loss, which down-weights easy examples and focuses on hard to classify ones, particularly beneficial for imbalanced datasets. Fourth, explore resampling techniques: oversampling minorities via SMOTE-like methods adapted for images (e.g., generating synthetic samples) or undersampling the majority class, though the latter risks losing valuable data. Ensemble methods, such as balanced bagging classifiers, could also be integrated by training multiple models on balanced subsets. Additionally, transfer learning from pre-trained CNNs (e.g., ResNet or EfficientNet adapted to grayscale) could provide robust initial features, reducing the impact of imbalance during fine-tuning. A two-stage training approach, first pre-training on a balanced subset, then fine-tuning on the full dataset might further enhance performance. Finally, incorporating multi-spectral or higher-resolution data from other Mars missions could enrich the feature space, potentially improving classification of subtle features. These enhancements would likely elevate minority class accuracies, making the model more viable for real-world planetary analysis.

5 References

- Doran, G. et al. (2020) *Mars orbital image (HiRISE) labeled data set version 3*. NASA Open Data Portal. Available at: <https://data.nasa.gov/dataset/mars-orbital-image-hirise-labeled-data-set-version-3> (Accessed: March 2026).
- Palafox, L. F. et al. (2017) ‘Automated detection of geological landforms on Mars using Convolutional Neural Networks’, *Icarus*, 301, pp. 120–127. doi: 10.1016/j.icarus.2017.09.026.
- Wagstaff, K. L. et al. (2018) ‘Deep Mars: CNN classification of Mars imagery for the PDS Imaging Atlas’, in *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1). doi: 10.1609/aaai.v32i1.11404.